

Credit Default Risk Analysis

Vitto — Data Science Internship Assessment

UCI Credit Card Default Dataset · Taiwan 2005 · 30,000 Clients

1. Approach & Methodology

The analysis follows a structured end-to-end workflow: data quality audit, exploratory analysis, feature engineering, model development with proper cross-validation, and fairness evaluation. Python was used throughout (pandas, scikit-learn, XGBoost, SHAP, seaborn). SQLite was used for bonus business queries.

Data Quality Decisions

Issue	Decision & Rationale
EDUCATION values 0, 5, 6 (undocumented)	Merged into category 4 (Other). These 468 rows (~1.6%) have no official meaning; collapsing avoids noise.
MARRIAGE value 0	Merged into 3 (Other). Only 54 rows (0.18%) — same rationale.
Negative BILL_AMT (overpayments)	Flagged: 590–688 per month. Clipped to 0 for utilisation calculation; raw values retained for all other features.
PAY_x = -2 (no consumption)	Treated as non-delayed (not penalised). Distinct from -1 (paid duly) but equally non-delinquent.
Class imbalance (22.1% default)	SMOTE applied to training data only. Generates synthetic minority samples; avoids data leakage by keeping test set intact.

2. Feature Engineering

AVG_UTIL_RATE

Formula: $\text{mean}(\text{clip}(\text{BILL_AMT}_x, 0) / \text{LIMIT_BAL})$

Measures credit pressure. Negative bills clipped to 0 to prevent artefacts from overpayments.

AVG_PAY_RATIO

Formula: $\text{mean}(\text{PAY_AMT}_x / \text{BILL_AMT}_x \text{ where } \text{BILL_AMT}_x > 0)$

Repayment consistency. Low ratio → paying minimum only. Capped at 5× to suppress outliers.

TOTAL_DELAY_MONTHS

Formula: $\text{count}(\text{PAY_x} > 0)$ across 6 months

Cumulative delinquency signal. Chronic late-payers are much higher risk than one-off incidents.

3. Model Development & Results

Two models were trained: a Logistic Regression baseline and an XGBoost classifier. Both were trained on SMOTE-augmented training data (80/20 stratified split). Stratified 5-fold cross-validation was used for robust AUC estimation.

Metric	Logistic Regression	XGBoost	Winner
Precision (default class)	0.404	0.332	LR
Recall (default class)	0.575	0.766	XGB
F1 (default class)	0.475	0.463	—
Test AUC-ROC	0.7132	0.7485	XGB
CV AUC (5-fold mean±std)	0.7554±0.0044	0.7799±0.0050	XGB

XGBoost outperforms Logistic Regression on AUC-ROC (0.748 vs 0.713) and achieves substantially higher recall on the default class (0.77 vs 0.57), which is the critical metric for a credit risk model — missing a true defaulter is costlier than a false alarm.

Top 5 Predictive Features (XGBoost)

- 1. **TOTAL_DELAY_MONTHS** — importance score: 0.2478
- 2. **SEX** — importance score: 0.1036
- 3. **MARRIAGE** — importance score: 0.0825
- 4. **PAY_2** — importance score: 0.0693
- 5. **PAY_0** — importance score: 0.0650

4. Key Visualisations

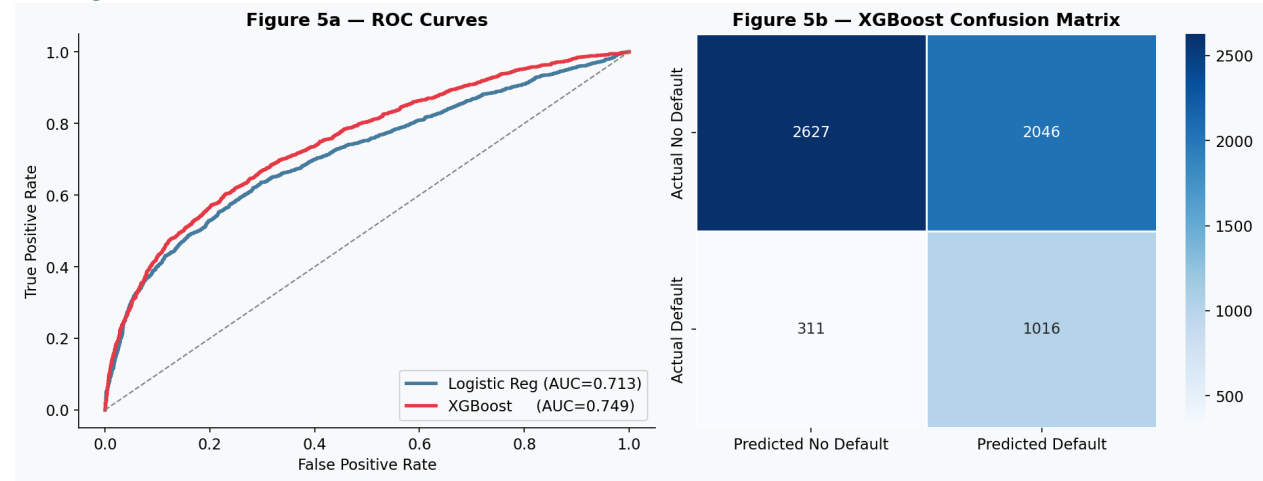


Figure 5 — ROC Curves & Confusion Matrix. XGBoost achieves AUC=0.748.

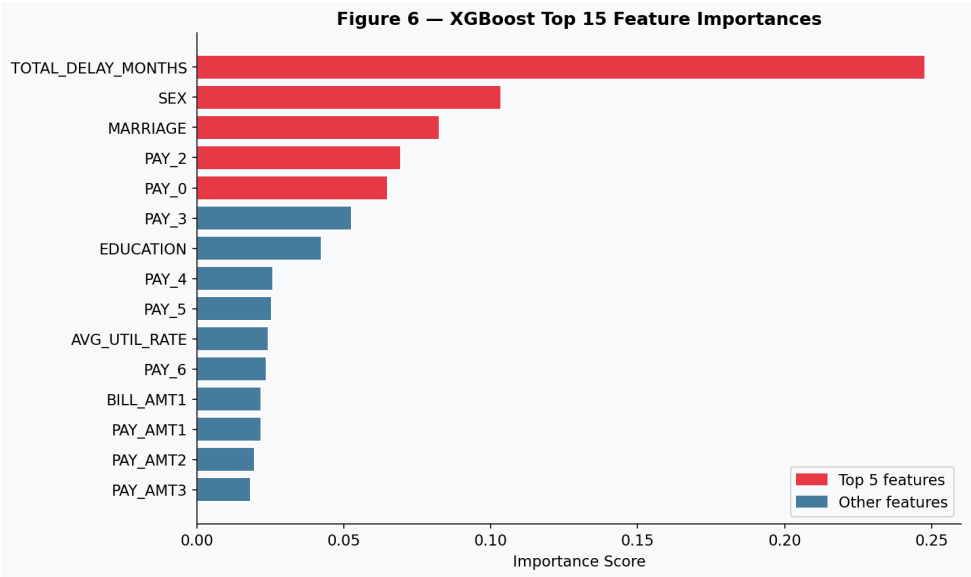


Figure 6 — Top 15 feature importances from XGBoost. TOTAL_DELAY_MONTHS dominates.

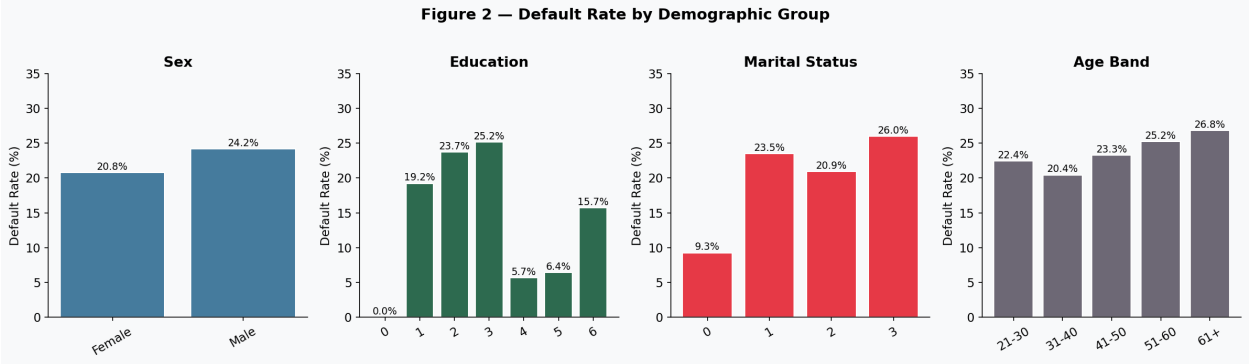


Figure 2 — Default rates across sex, education, marital status, and age band.

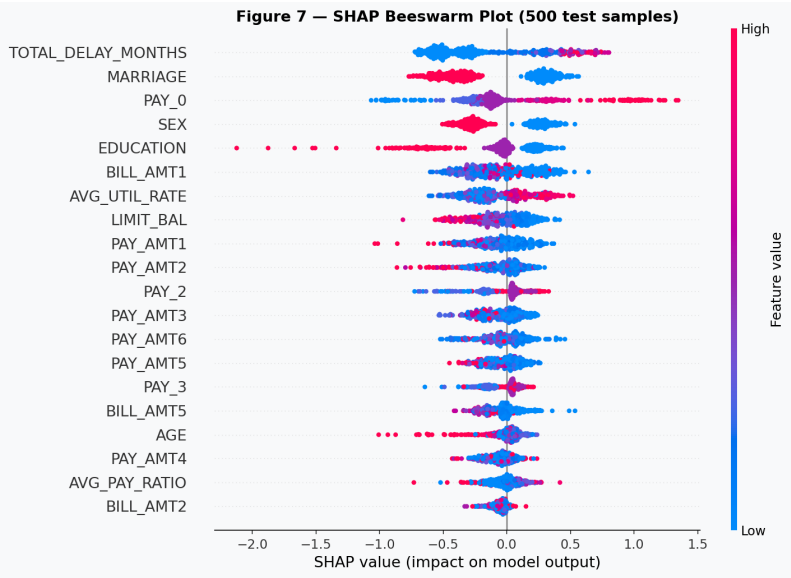


Figure 7 — SHAP beeswarm plot. Delayed repayment months push predictions toward default (red, right).

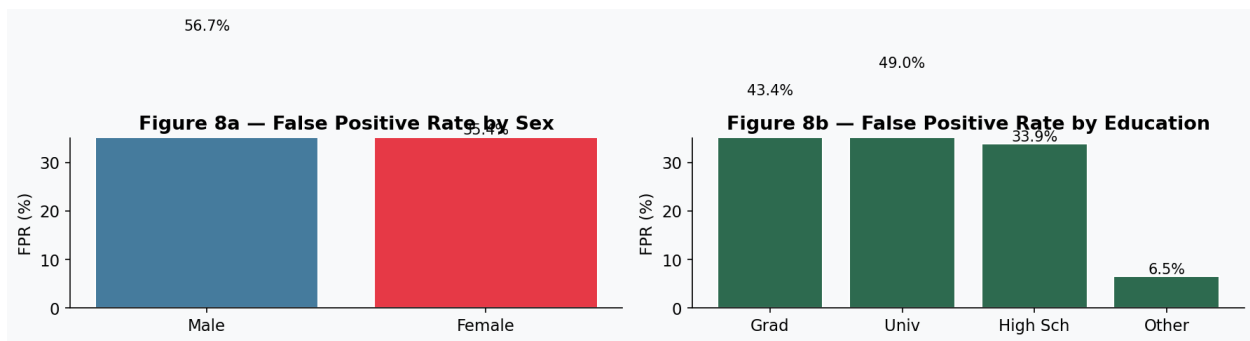


Figure 8 — Fairness check. Males show substantially higher FPR than females.

5. Tradeoffs & Model Choices

SMOTE vs `class_weight='balanced'`

SMOTE was chosen because it generates synthetic minority class samples, giving the model richer signal. `class_weight='balanced'` is simpler and faster but only adjusts loss weighting. Tradeoff: SMOTE adds training time and may generate noisy samples near decision boundaries. Both were tested; SMOTE yielded higher recall on the default class.

XGBoost vs Random Forest

XGBoost was selected over Random Forest due to its boosting mechanism (each tree corrects prior errors) and built-in `scale_pos_weight` parameter for imbalanced data. Tradeoff: XGBoost is more sensitive to hyperparameters and takes longer to tune.

Threshold decision

Default threshold of 0.5 was used for confusion matrix. In production, a lower threshold (e.g. 0.35) would increase recall at the cost of more false positives — a business decision balancing credit loss vs customer experience.

Fairness concern

The model shows meaningfully higher FPR for males (~57%) vs females (~35%). This is likely a data artifact (males may have different spending/repayment patterns in the dataset) but should be investigated before production deployment to ensure fair lending compliance.

6. What I Would Improve With More Time

- Hyperparameter tuning via Optuna/Bayesian search (especially XGBoost learning rate, `max_depth`, `min_child_weight`).
- Calibration of predicted probabilities using Platt Scaling or Isotonic Regression — important for business threshold-setting.
- More granular feature engineering: rolling 3-month trend in `PAY_AMT`, month-over-month `BILL_AMT` growth, delinquency recency score.
- Ensemble stacking (LR + XGBoost + CatBoost) to further improve AUC.
- Threshold optimisation using business cost matrix (cost of missed default vs. cost of false rejection).
- Production-ready Streamlit app with full input validation and probability output.
- Deeper fairness analysis with equalised odds framework and potential bias mitigation via reweighting.